

Applications of Physical AI Reflex Modules

Energy harvesting and battery balancing

PV and battery reflex-control opportunity

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INTRODUCTION

This work presents a novel architecture for local, ADC-light reflex control.

Binary spintronic crossbars are used here not as AI accelerators (Policy-Layer), but as **Physical AI Reflex-Layer modules**.

The same principle applies across a wide spectrum of energy systems:

- ❑ **Photovoltaic (PV)**: fast bypass and MPPT stabilization
- ❑ **Battery balancing**: low-dissipation energy routing
- ❑ **Thermoelectric (TE) and thermophotovoltaic (TPV)** – mismatch compensation
- ❑ **Piezoelectric harvesters**: local vibration-response adaptation
- ❑ **Beyond energy**: robotics, IoT, and autonomous systems

We wil present a new system architecture and new application domains for physical devices.

SPIN-PV: Presentation Agenda

1. **CONCEPTUAL FOUNDATION** : Mirror metaphor
2. **SPINTRONIC BINARY CROSSBAR**: What it is and what it is missing
3. **REFLEX Layer and POLICY Layer in Physical AI**
4. **CLASSICAL PV AND BMS LIMITATION**
5. **PV-BATTERY REFLEX BALANCE EXAMPLE**
6. **BENEFITS**: Efficiency, heat, latency, and wiring
7. **SCALING**: Beyond PV, TE-TPV, ..IoT, Robotics, EROIE/EROI
8. **CONCLUSIONS**: Crossbar as a Physical AI Reflex-Layer Module

A mirror reflects without thinking

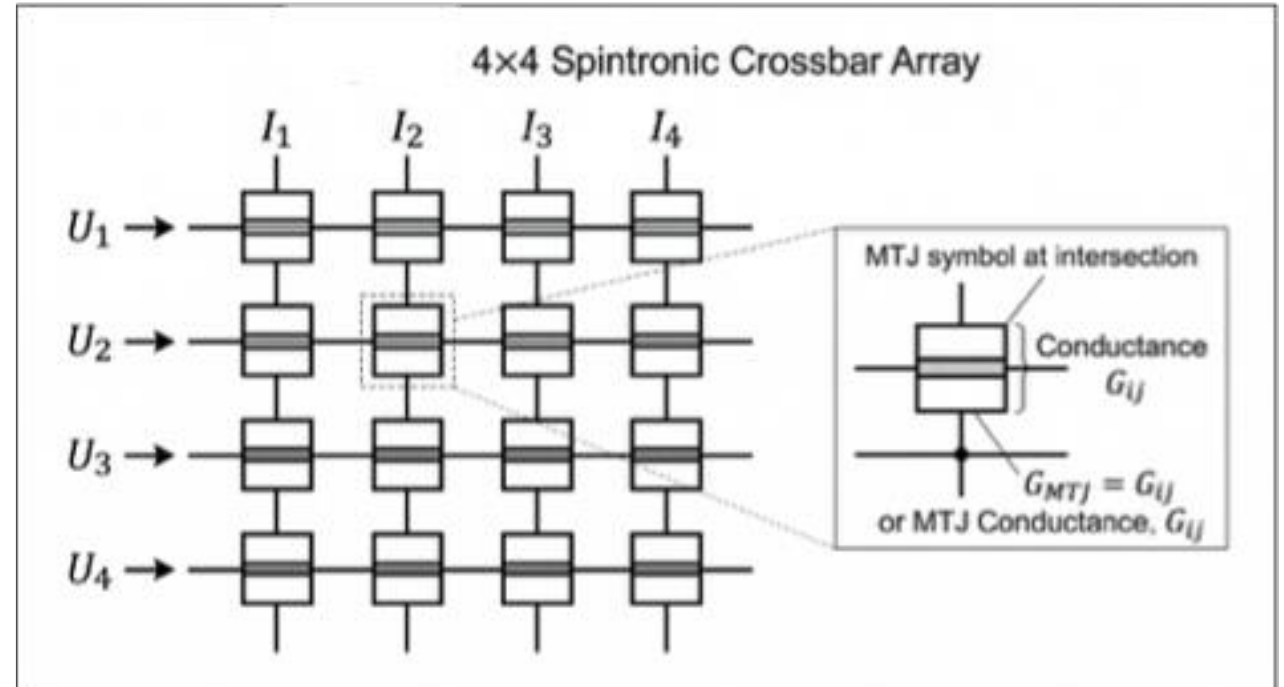
From mirror to binary spintronic crossbar: A minimal metaphor for a local physical reflex

- The mirror has no processor.
 - It runs no algorithm.
 - It does not recognise the image.
 - Yet it gives an ordered response.
-
- The structure of the material performs the transformation.
 - A physical system can transform an input into an output without a digital computation.
 - The binary spintronic crossbar is the engineering candidate for fast local events.
 - Low-latency physical reflex can beat sequential sense, compute, act control.

What is a binary spintronic crossbar?

A programmable decision grid with magnetic binary intersections

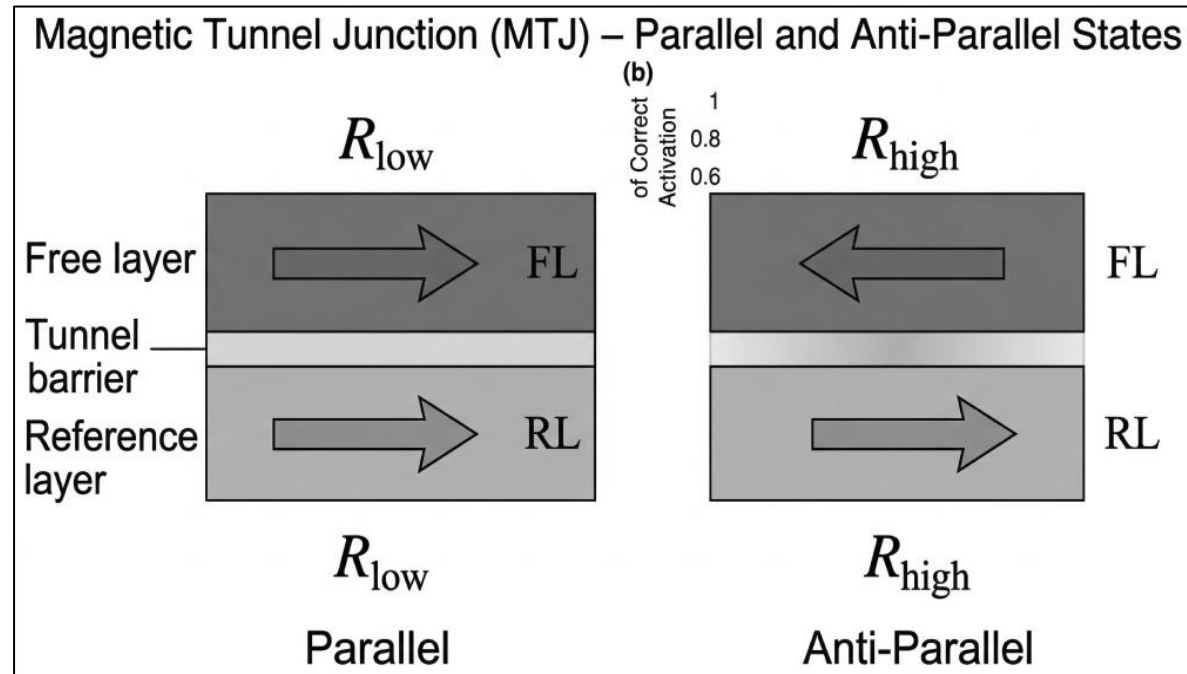
- A crossbar is a programmable decision grid.
- Rows carry input conditions.
- Columns collect outputs or actions.
- At each crossing there is a programmable element.
- In a spintronic crossbar this element can be an MTJ.
- Binary state: strong or weak connection.
- The grid stores a rule map.
- **Local conditions enter; local actions leave**



*Classical convention: horizontal inputs, vertical outputs,
MTJ conductance at each intersection.*

What makes the crossing spintronic?

The MTJ stores a non-volatile binary conductance state



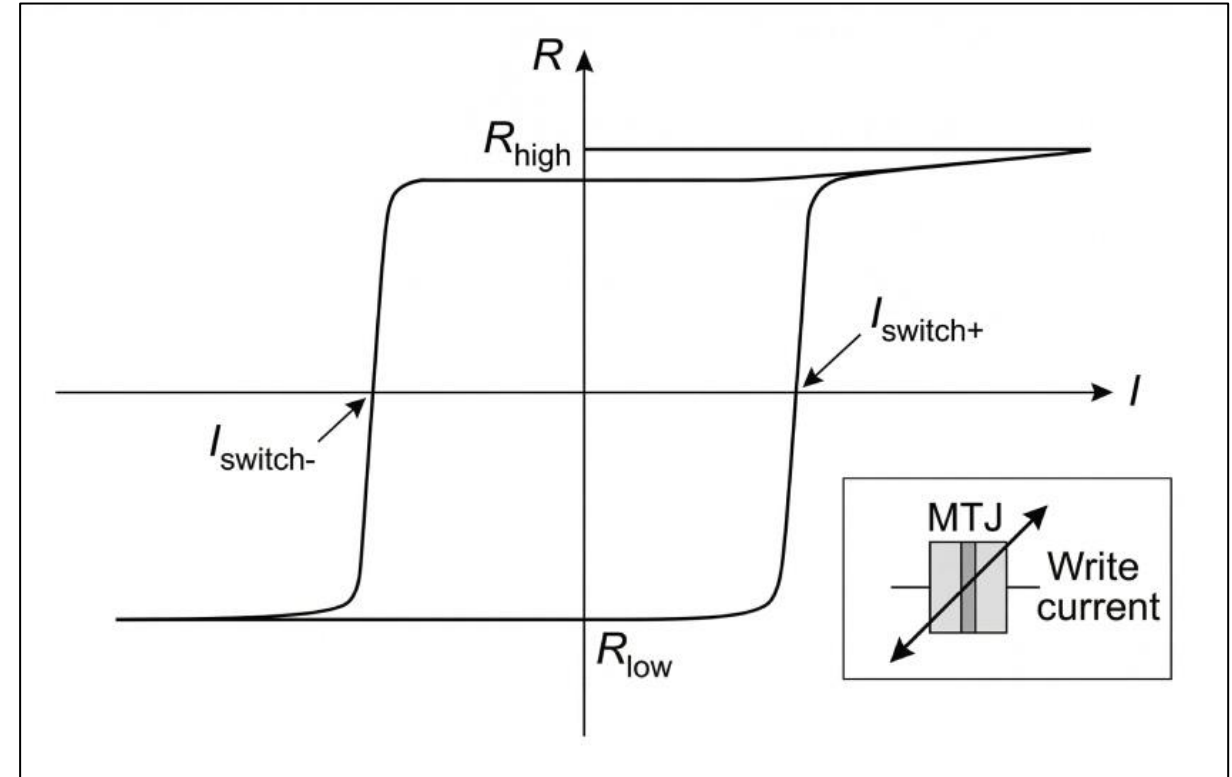
MTJ principle: two magnetic states create two resistance states

- A magnetic tunnel junction has:
 - a reference magnetic layer
 - a tunnel barrier
 - a free magnetic layer
- **Parallel magnetisation gives low resistance.**
- **Anti-parallel magnetisation gives high resistance.**
- **The MTJ at the cross-point stores a non-volatile binary state.**
- **The crossbar “waits” with the rule already programmed.**

Programming and retention

The rule remains stored without continuous power

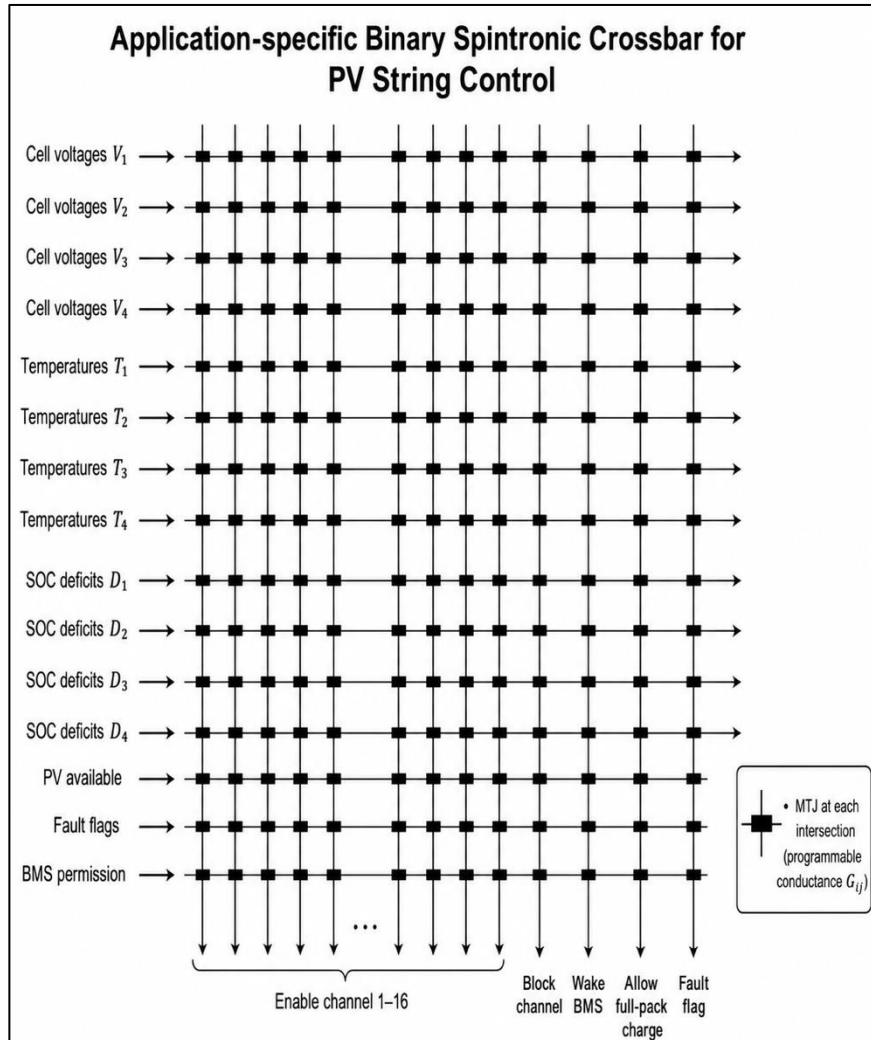
- A write pulse can switch the MTJ between two stable resistance states.
- One state represents a strong connection.
- The other state represents a weak connection.
- After programming, the state is retained magnetically.
- This is why the crossbar can wait with very low standby power.
- When a local event arrives, the response is already physically prepared.



Binary switching: the device moves between R_{low} and R_{high} after a sufficient write pulse

How the crossbar operates as a local rule Module

Application-specific example: PV-side control



- **Rows bring local conditions:**
cell voltages, temperatures, source availability, fault flags, BMS permission.
- **Columns send local actions:**
enable channel, block channel, wake BMS, report fault.
- Programmed MTJ intersections encode the rules.
- Condition enters.
- Action leaves.

Large NxM can avoid multiplexing in time-critical reflex path

Decision path versus power path

The crossbar commands; the power electronics carry energy

Decision path

- sensors → local flags → crossbar rule module → gate-enable / block / wake commands
- **Only small sensing and decision currents are present.**

Power path

- PV / auxiliary source / battery bus → DC-DC converter → MOSFETs → bypass path or isolated injection channel
- **The real energy current flows outside the crossbar**

Analogy: the crossbar is like a **pilot valve or a railway switch. Small signal selects the route; main circuit carries the energy.**

What exists today — and what is still missing

Technology exists; the Physical AI Reflex Architecture does not

What exists today

- Commercial MRAM products.
- Embedded MRAM foundry routes.
- Examples **outside Europe** include: Everspin, Avalanche Technology, Samsung Foundry, GlobalFoundries, UMC with Avalanche route, TSMC research and technology routes, and Numem.
- Manufacturable Binary magnetic arrays.

What is missing for our purpose

- Today's products are mainly memory products, embedded memory macros, computing support, or research-oriented arrays.
- They are designed to store, retrieve, or accelerate data operations.
- **We are not looking for another AI accelerator**
- **We need a Physical AI Reflex-Layer Module:** local physical states in, protective / routing / control actions out, ADC-light, close to the physical event.

MRAM proves the material route; it does not yet provide the reflex architecture

Reflex Layer and Policy Layer

Physical AI as seen from nature

- Physical AI must act in the real world under limits of energy, latency and robustness.
- Nature does not use one central processor.
- Urgent local events are handled by Reflex-Layers Modules.
- Learning, planning and optimisation are handled by Policy layers.
- Design principle: use only the inference level that the event really needs.

Reflex Layer

fast, local, ADC-light.
Binary crossbars

Policy Layer

slower, richer, learning.

**Energy-proportional intelligence:
not every physical event deserves the same
computation.**

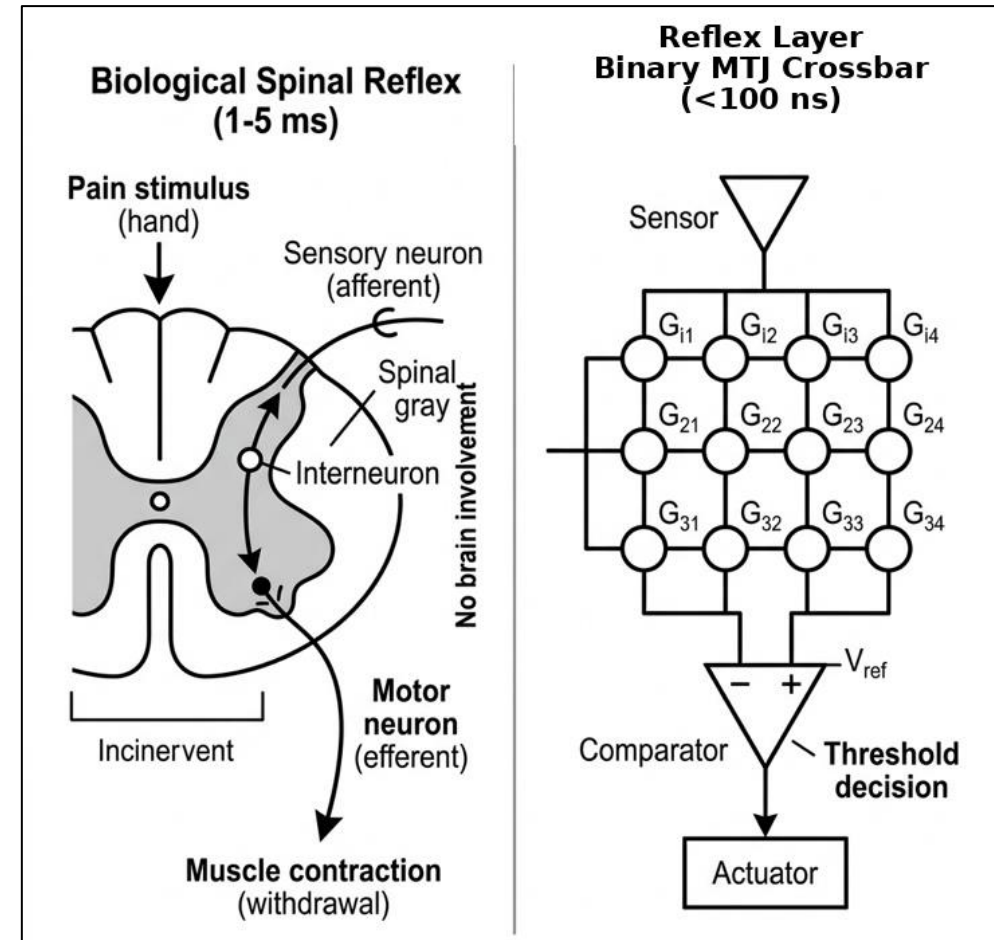
We deliberately start with **binary** because most of the critical decisions in energy harvesting and battery management are naturally binary or low-resolution !!

Reflex: In everyday language we might call this instinct. More precisely, in biology and engineering, we call it a reflex: a fast, local, automatic response that acts before full deliberation.

Nature: reflex before central computation

Spinal reflex as ADC-light physical intelligence

- A spinal reflex does not wait for cortical interpretation.
- The local circuit receives a stimulus and triggers a protective action.
- Electronic analogy:
sensor $\rightarrow$ binary rule map $\rightarrow$ comparator $\rightarrow$ actuator.
- No high-resolution ADC is required in the urgent path.
- The Policy Layer may tune thresholds later, **but the first response is local.**



Why the spintronic binary crossbar is important

Emulating nature through simple physical rules

Not mainly an in-memory-computing accelerator.

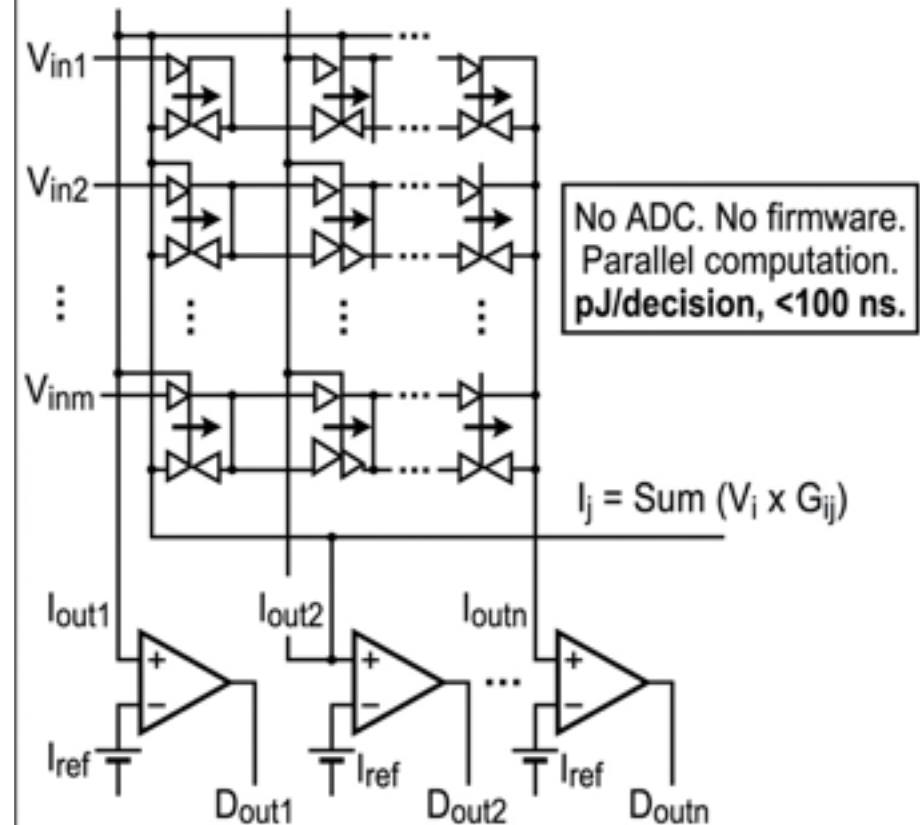
Not high-precision AI inference

Binary MTJ states store local rules

Rules are non-volatile and physically prepared.

Parallel low-resolution rule execution replaces slow sequential polling.

Binary MTJ Crossbar as Reflex-Layer



Simplicity as the last sophistication: the reflex is simple because it is already physically prepared. Doing simpler things, at the right place, can do much more.

Reflex Layer versus Policy Layer

The energy-latency trade-off

- The goal is not to replace advanced AI
- The goal is to place each decision at the right level.
- Reflex layers handle repetitive, urgent, local decisions.
- Policy layers handle learning, prediction, action selection and global context.
- Higher inference means complexity, energy and latency also increase.

Policy Layer

learning / prediction / global context

Local Policy

action selection / optimisation

Digital Reflex

deterministic routines

Analog Reflex

thresholds / timers

Physical Reflex

binary local rule
execution

Lowest sufficient Layer acts first

How layers work together

Fast execution below, slower learning above

- The layers are concurrent, not sequential.
- Upward flow is compressed: events, faults, status and measured response.
- Downward flow is configurational: thresholds, gains, permissions and rule maps.
- Policy layer learns and configures
- Reflex layer executes local actions

Policy Layer

learn, optimise, configure

Reflex Layer

execute local actions

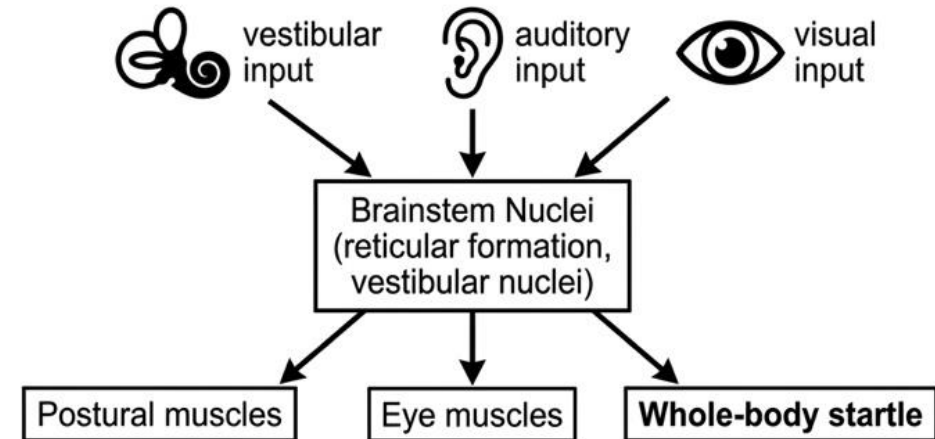
If the higher layer is slow or unavailable, the local reflex layer still protects the system.

Second example from nature: brainstem and CPG

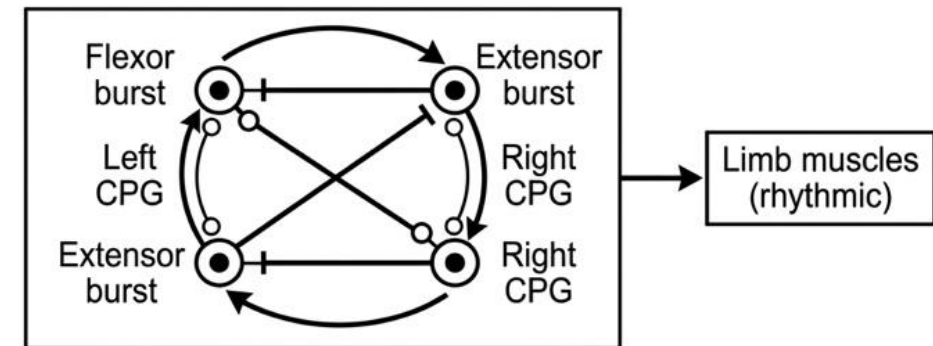
Local coordination without full deliberation

- Brainstem circuits integrate vestibular, auditory and visual inputs rapidly.
- Central pattern generators produce rhythmic motor patterns without cortical command.
- This is still local physical intelligence: repeated actions are generated close to the body.
- Engineering lesson: simple local circuits can coordinate more than a single emergency reflex.
- Crossbar analogy helps emulate this. many simple inputs, stored rules, local outputs.
- Local simplicity can coordinate complex behaviour

Brainstem Reflex Integration (10-50 ms)



Central Pattern Generator (CPG) for Locomotion



Classical PV and BMS limitations

System integration problem — MPPT — energy losses

Before introducing the binary spintronic crossbar, we first identify where energy is lost in classical architectures.

Capture → Condition → Route → Store → Protect → Use

PV side

The source is distributed and dynamic. Partial shading and mismatch move the operating point before energy reaches storage.

Battery side

Cells age differently. Classical balancing either burns useful energy as heat or adds converter complexity.

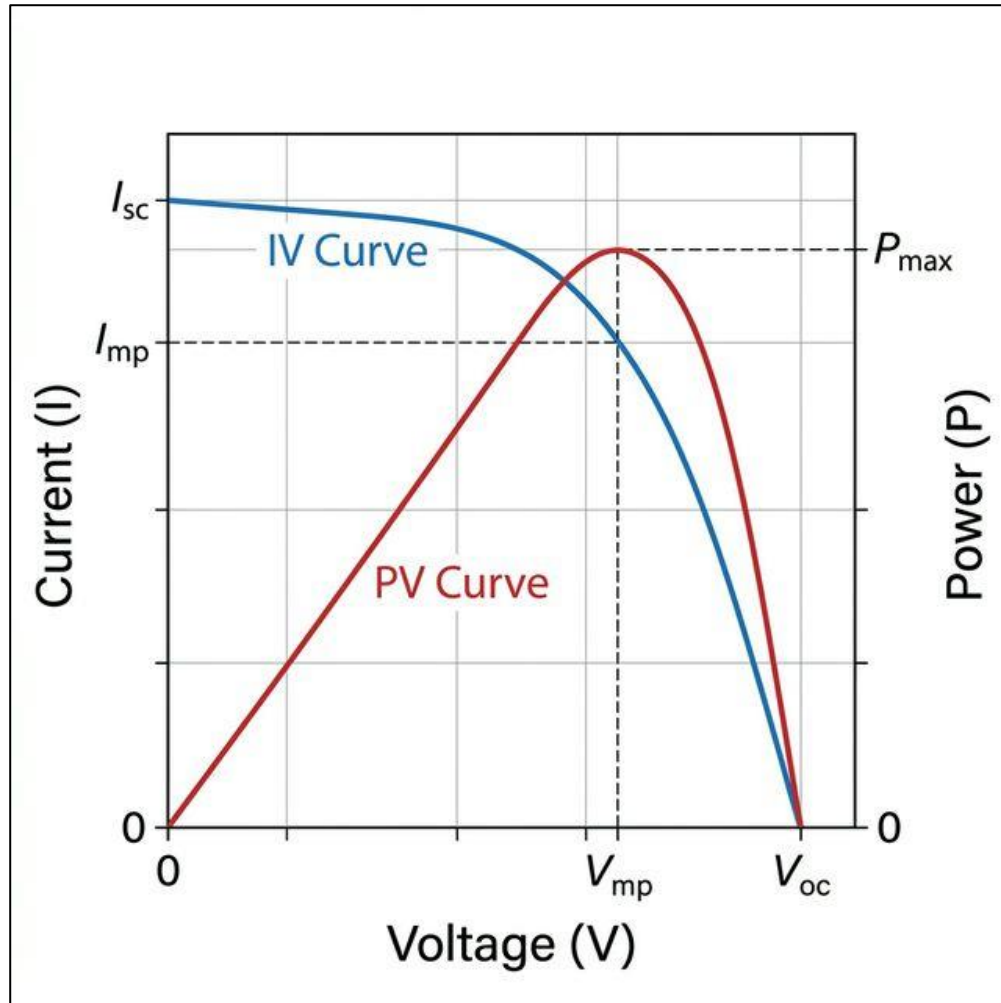
System level

Efficiency is not only a component value. It is the result of sensing, routing, conversion, storage and protection.

Harvested energy can be lost before storage, and stored energy can be wasted during balancing.

Classical PV harvesting: the moving maximum

MPPT is needed because the PV sweet spot shifts with light, temperature and load



PV behaviour

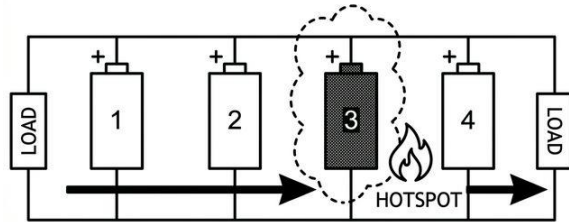
- A PV cell has an I–V curve and a P–V curve.
- Only one point gives maximum power under uniform illumination.
- The maximum point moves with sunlight, shadows, temperature and load.
- A fixed load rarely keeps the PV string at maximum power.

Classical answer

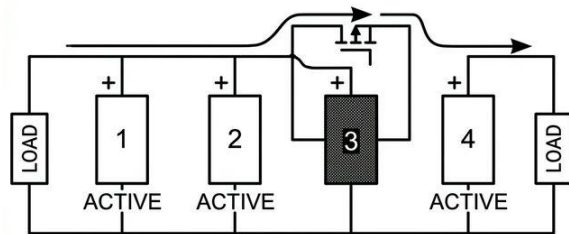
Classical MPPT works best when the generator is uniform and changes slowly.

Partial shading breaks the simple PV picture

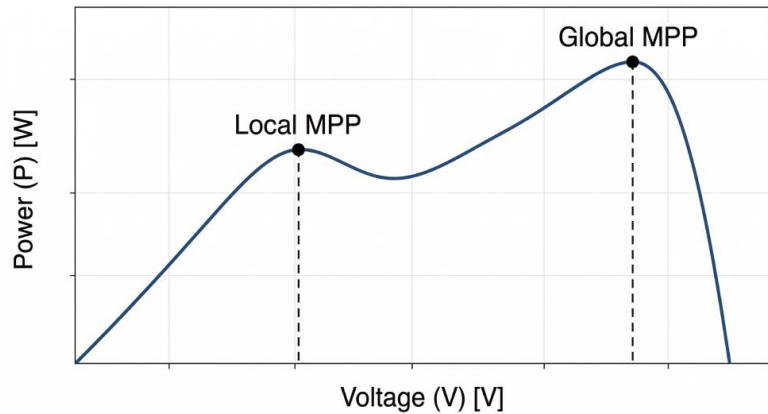
One weak section can limit the whole string and create multiple power maxima



(a)



(b)



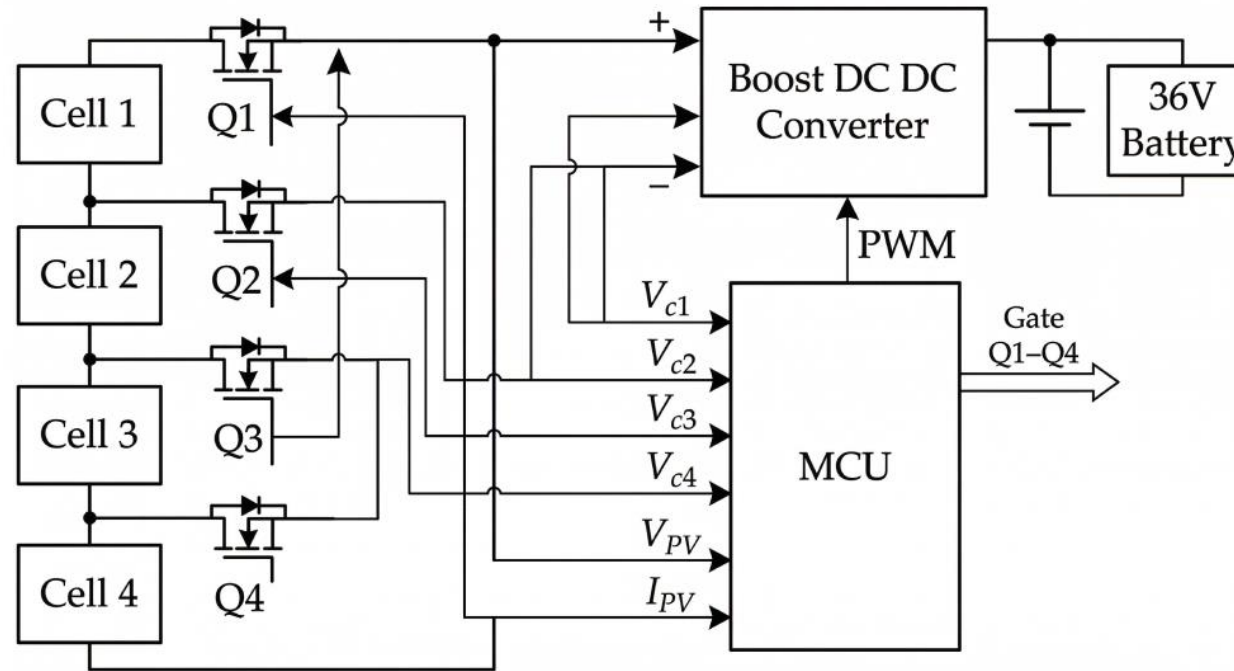
What changes under shading

- In a series string, the same current must pass through every cell or substring.
- **A shaded element becomes the bottleneck** and can create reverse-bias or hot-spot risk.
- Bypass helps, but shading is local and fast
- Partial shading can create local and global maxima.

The PV source seen by the converter is no longer smooth, slow or globally obvious.

Classical PV harvesting chain

PV string, bypass switches, boost converter and battery are part of one control loop

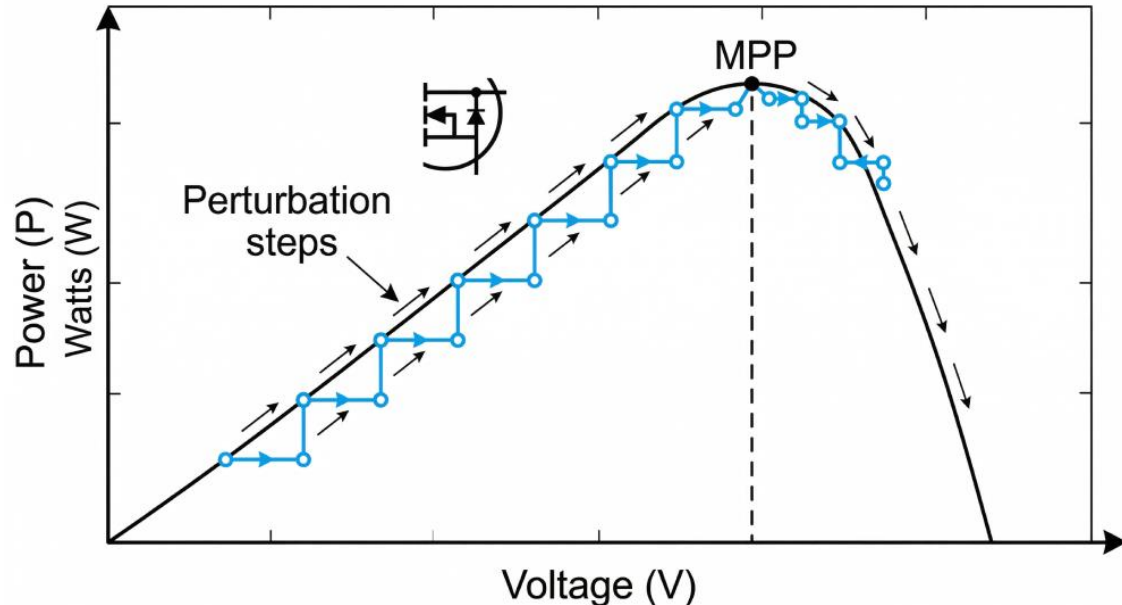


Block diagram of the MPPT system. each cell has a MOSFET bypass switch to redirect current around a shaded cell and recover part of the available energy. The MCU measures string voltage and current. The MCU controls the boost converter via a PWM signal (duty cycle D) and drives the bypass MOSFETs. The microcontroller that samples voltages and currents, runs the MPPT algorithm, The boost converter interfaces the PV string with the battery, its output charges the battery.

MPPT is not isolated software. It depends on the whole chain: cell states, bypass paths, converter dynamics, battery acceptance and control latency

Classical MPPT sees the aggregate, not the local cause

The controller reacts to total PV voltage and current after the source has already changed



Perturb and Observe: perturb duty cycle → measure power → keep or reverse direction

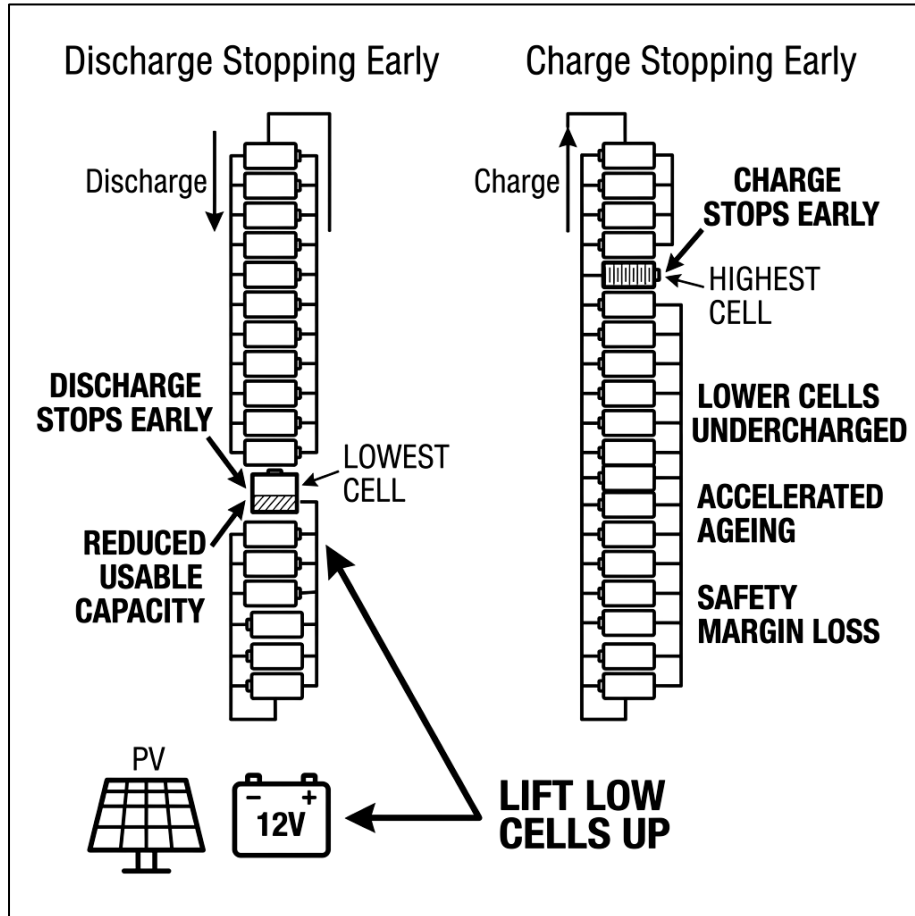
Limitation under fast shading

- The panel is treated as one source at the converter input.
- The controller measures total voltage and total current, not local cell conditions.
- Under fast partial shading, the P–V curve changes while the algorithm is still searching.
- Useful energy is lost during every transient and during every wrong local-maximum search.

Classical MPPT remains essential but it sees the aggregate, not the local cause

Battery packs are governed by the weakest cell

Balancing is not optional: it protects capacity, lifetime and safety



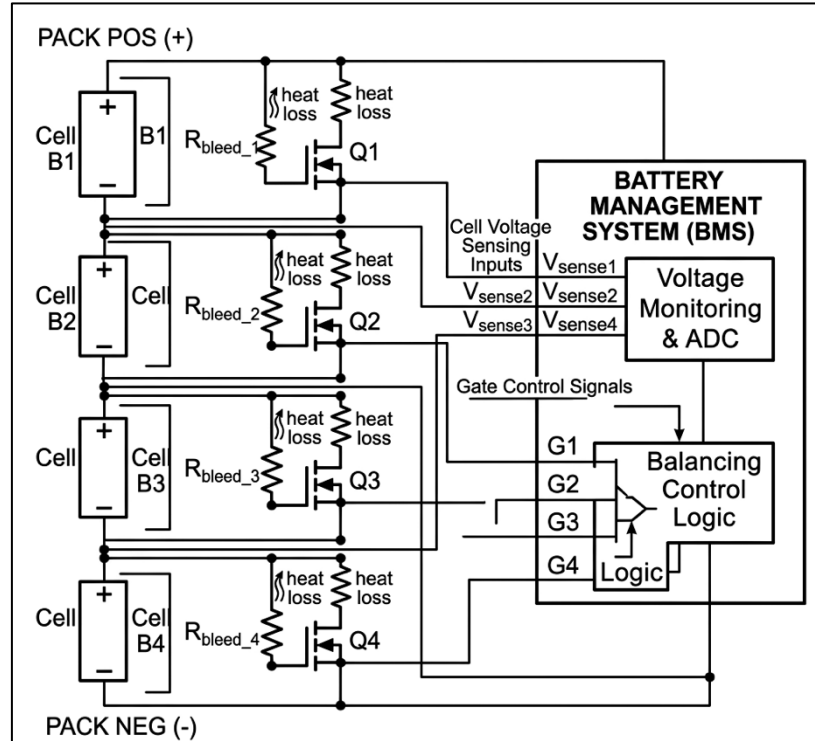
Series-pack consequence

- The same current flows through all cells or clusters.
- During discharge, the lowest cell reaches its limit first.
- During charge, the highest cell reaches its limit first.
- Without balancing, usable capacity shrinks and ageing accelerates.

The challenge is correcting imbalance without wasting useful energy.

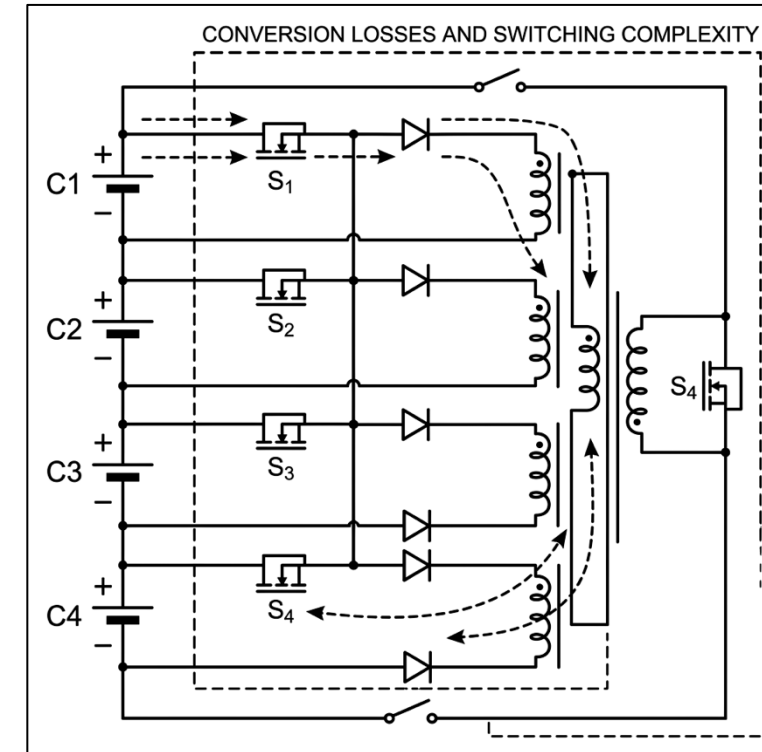
Classical balancing: heat or complexity

Passive balancing is simple but dissipative; active balancing saves heat but adds hardware and control burden



Passive Balancing

Robust and cheap, but useful energy becomes heat. Balancing speed is limited by thermal constraints. Higher cells are discharged through resistors



Active Balancing

Lower direct heat loss, but more switches, converters, EMI, firmware and failure modes. Energy is moved through capacitive, inductive or converter paths

System-level summary: where the classical chain loses energy

The need for local binary reflex modules on the PV side and battery side

PV limitation

Global MPPT sees total output. Partial shading is local and fast.

BMS limitation

Balancing is necessary, but classical correction is slow, hot or complex.

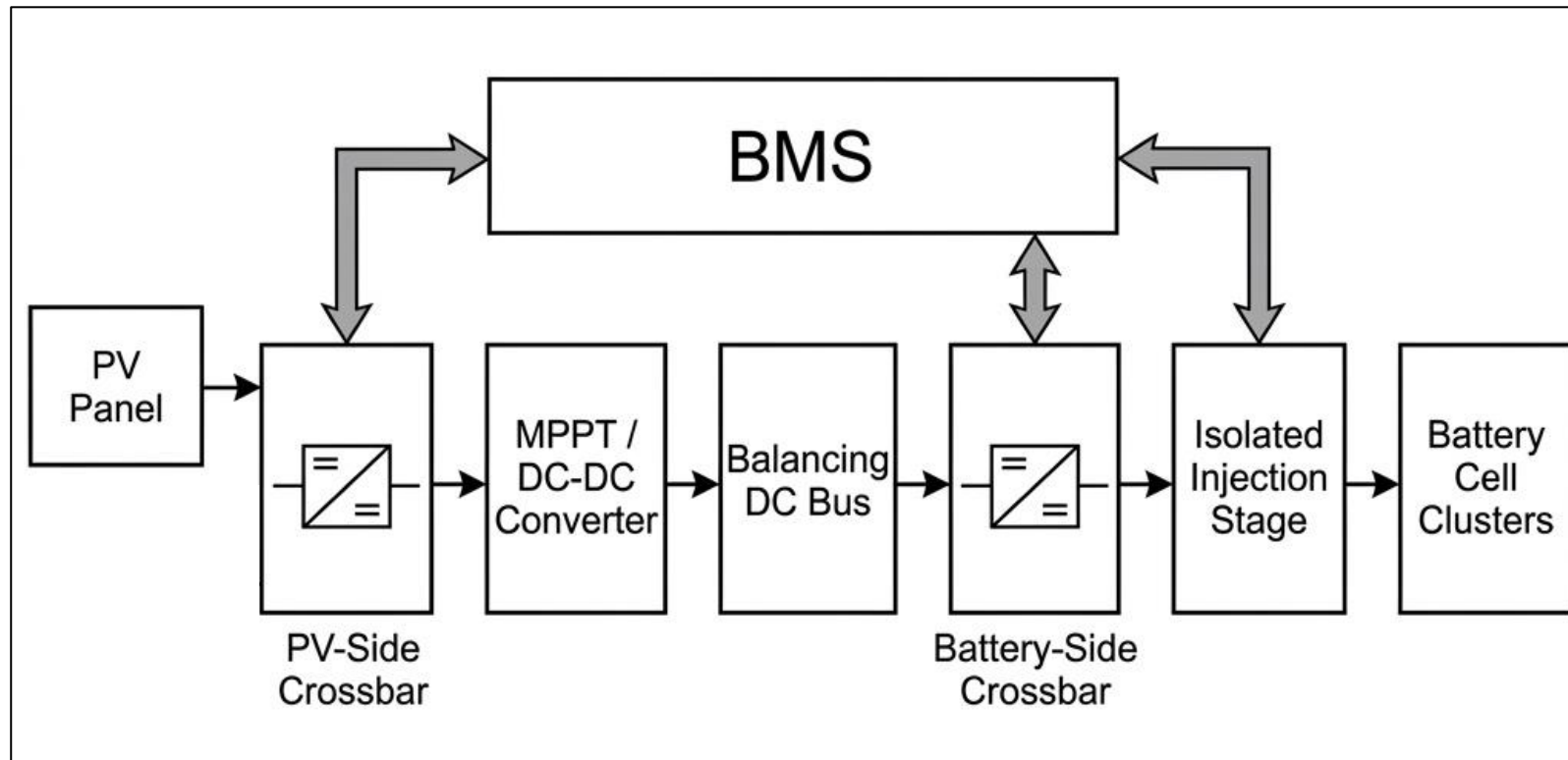
Required direction

- Sense local physical events closer to the source.
- Convert them into low-resolution binary flags.
- Use a local Reflex Layer Block to command bypass, routing, wake-up or inhibition.
- Let classical MPPT and BMS remain the policy and safety supervisors.

The binary spintronic crossbar is introduced as local, ADC-light reflex module.

PV-Battery Reflex Balance Example at a Glance

Novel cooperative binary crossbar architecture



PV side captures more usable energy

Battery side routes energy where it is needed

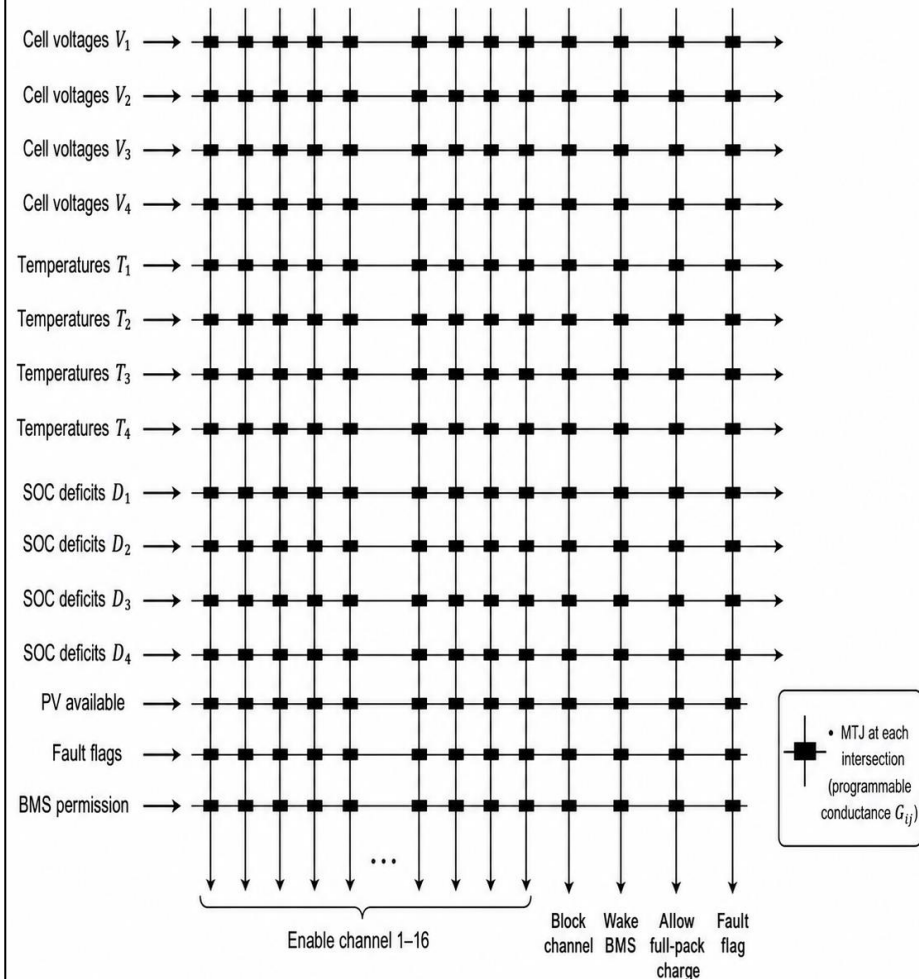
Main BMS remains safety and policy supervisor

The aim is not to replace MPPT or BMS, but to add ADC-light local reflex modules

PV-side need: react before energy is lost

Partial shading is local and fast; classical MPPT sees the combined result later

Application-specific Binary Spintronic Crossbar for PV String Control



- Shading degrades a substring before the system-level controller reacts.
- Input: PV-side crossbar module receives local flags: voltage, current, temperature, shading risk.
- Output: commands fast bypass, reconnection, inhibition or MPPT wake-up.
- The crossbar is a decision path; MOSFETs and converters carry the power path.

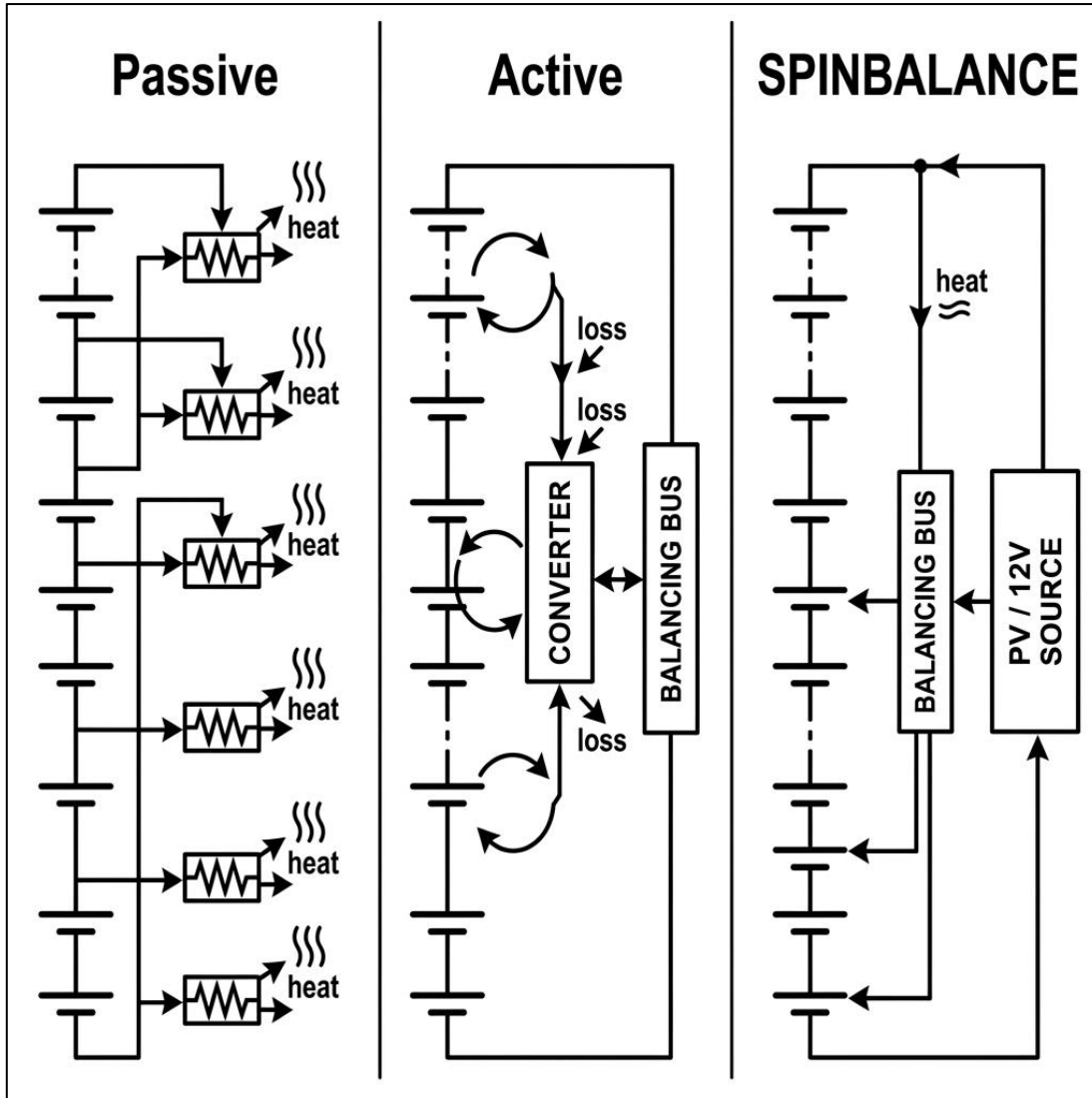
Local physical event → binary flags → immediate local command

The PV-side module improves the source that the MPPT converter sees.

Large NxM can avoid multiplexing in time-critical reflex path

Battery-side need: lift low cells instead of burning high cells

Balancing should recover useful capacity without turning energy into heat

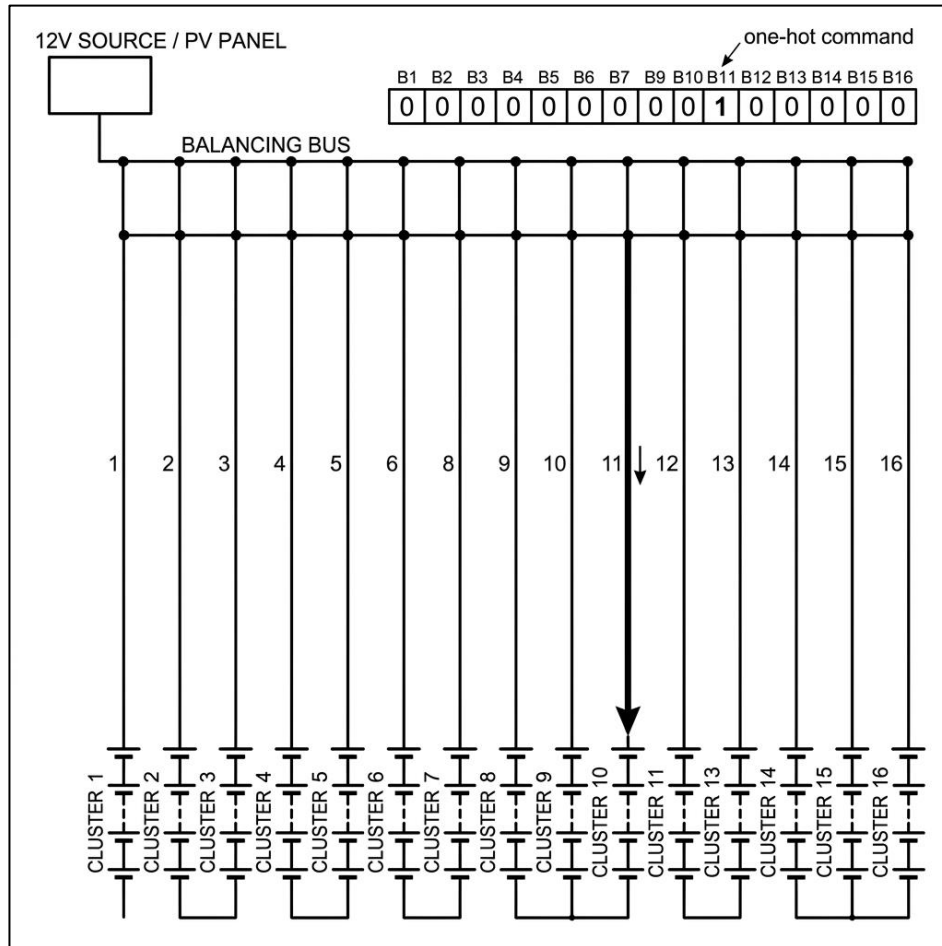


- Passive balancing removes energy from high cells and dissipates it as heat.
- Active balancing moves energy but add complexity
- SPIN-BALANCE uses PV or auxiliary energy to lift the weakest safe cells.

Less heat • faster correction • more useful stored energy

Battery-side crossbar as a fast energy-route selector

One controlled source can be directed to the lowest safe cluster

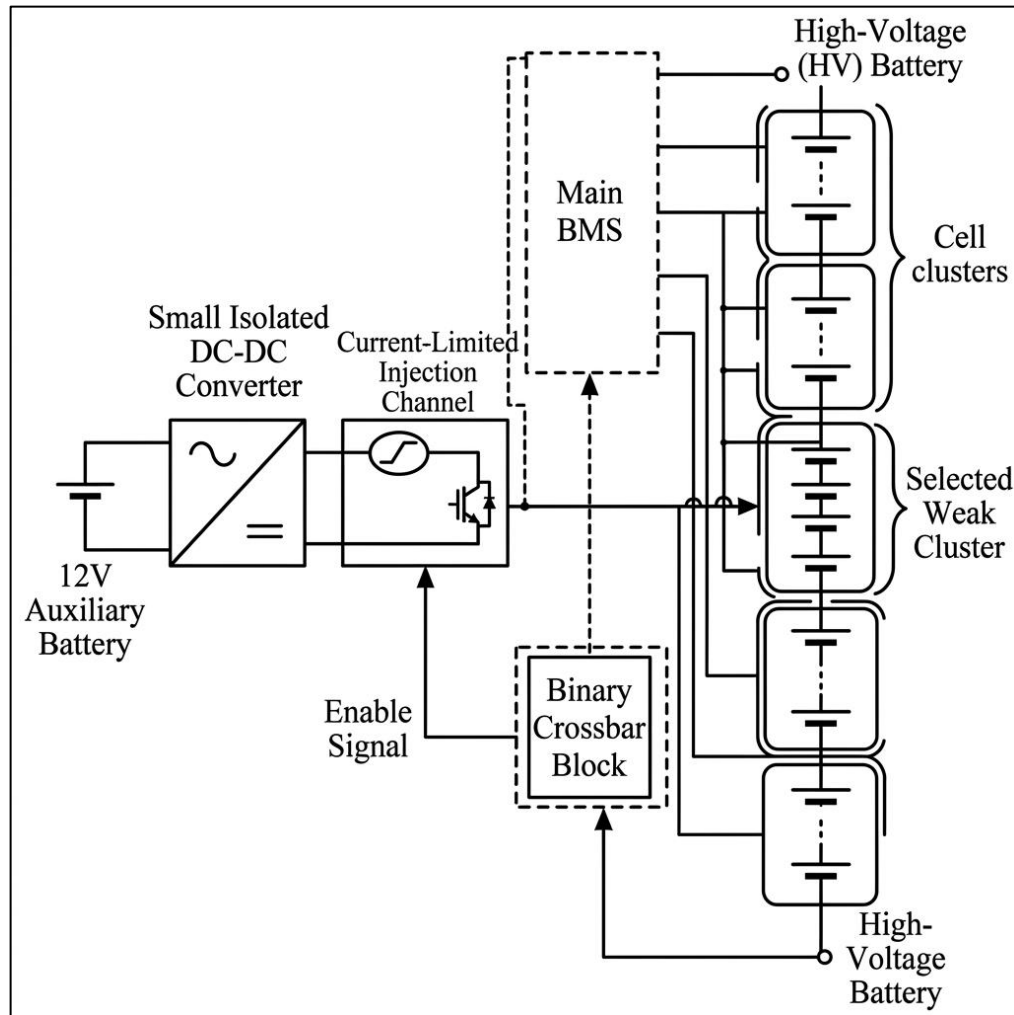


- **Inputs** are low-resolution flags: low cluster, temperature safe, voltage safe, source available, BMS permission, no local fault.
- **Outputs** are channel commands: enable selected injection channel, block unsafe channel, wake BMS or signal fault.
- **Operation mode:** One-hot mode keeps only one injection channel active: simple, safe, certifiable and easy to debug.

The crossbar selects the path; power electronics carry the energy.

Battery balancing can use PV or an auxiliary source

External or auxiliary energy is conditioned into a balancing bus, then routed locally



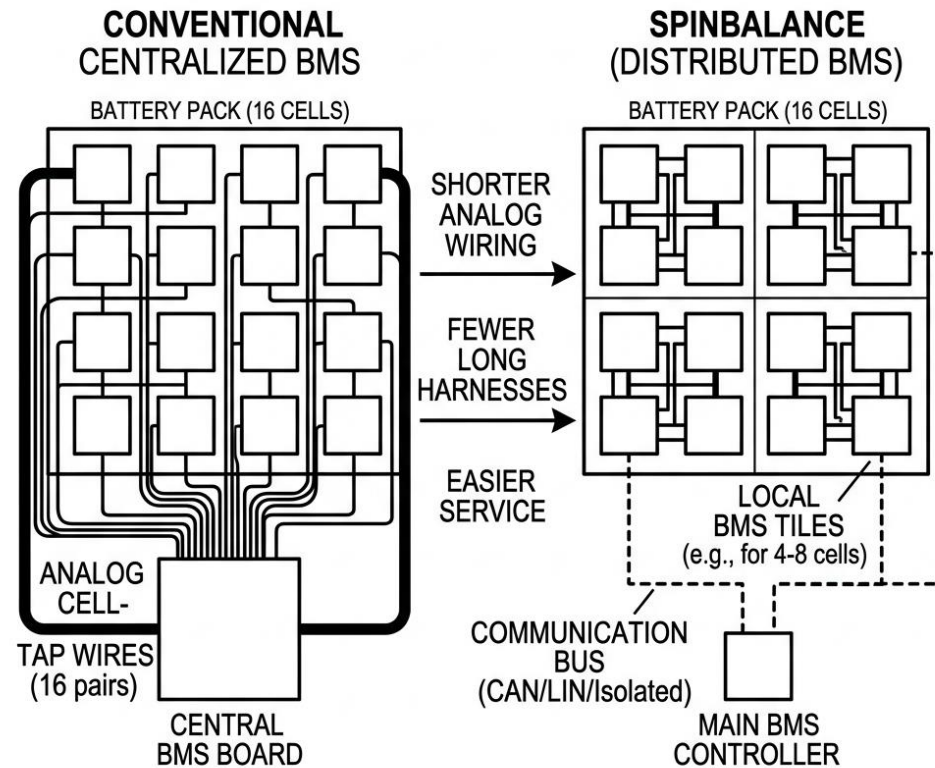
- PV energy, 12 V auxiliary battery, 24/48 V service rail or charger can feed the balancing bus.
- The battery-side crossbar module selects the eligible weak safe cluster.
- Local modules reduce the need to centralize every low-level balancing reflex.
- Short local cell connections replace many long analog balancing-related runs.
- Main BMS still supervises thresholds, safety, logging and fallback.

Simplicity advantage: local decision, short wiring, controlled injection

Why the battery-side crossbar improves wiring, energy and time

Local reflex modules reduce long analog runs and accelerate balancing correction

WIRING SIMPLIFICATION: CENTRALIZED vs. DISTRIBUTED BMS



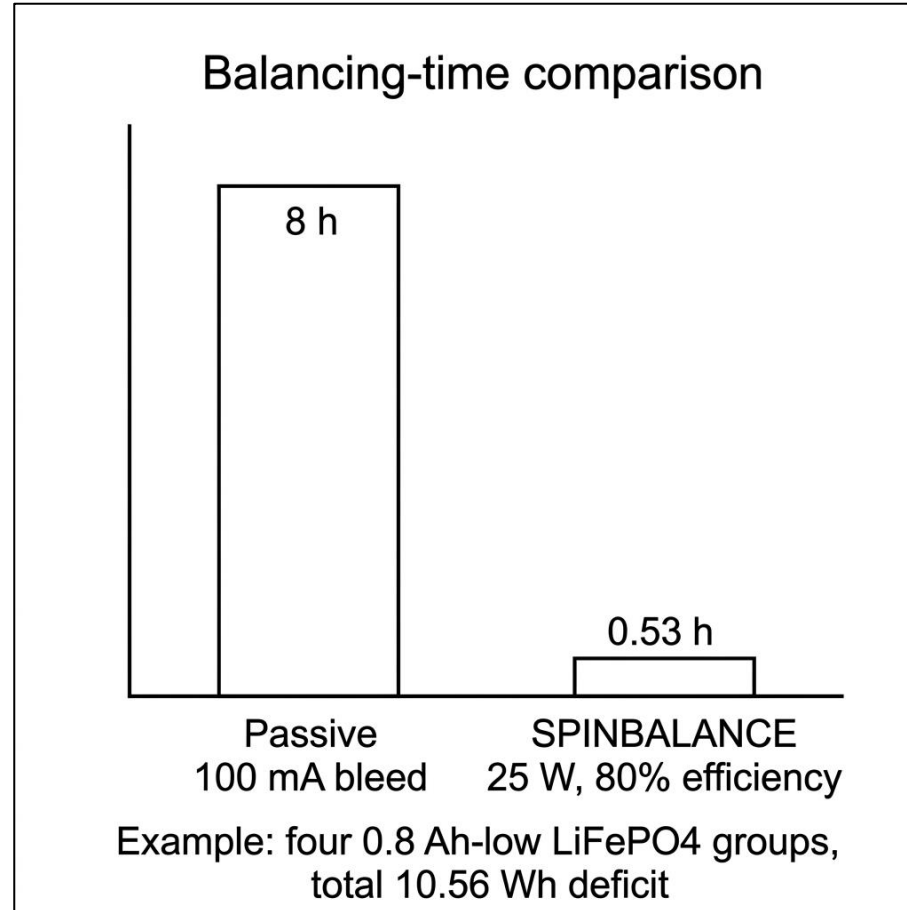
Wiring simplification: centralized versus distributed BMS/SPIN-BALANCE blocks

Classical balancing	SPIN-BALANCE direction
Many long analog cell-tap wires	Short local cell wiring inside modules
Balancing heat on central boards	External/auxiliary energy lifts low cells
Slow correction limited by bleed current	Higher useful injection power can shorten balancing
Centralized decision overhead	Local crossbar executes repetitive routing
More connectors, noise pickup and service complexity	Lower wiring burden and easier modular replacement

The BMS remains the safety authority; the crossbar module decentralises the repetitive balancing reflex.

Why the battery-side crossbar improves wiring, energy and time

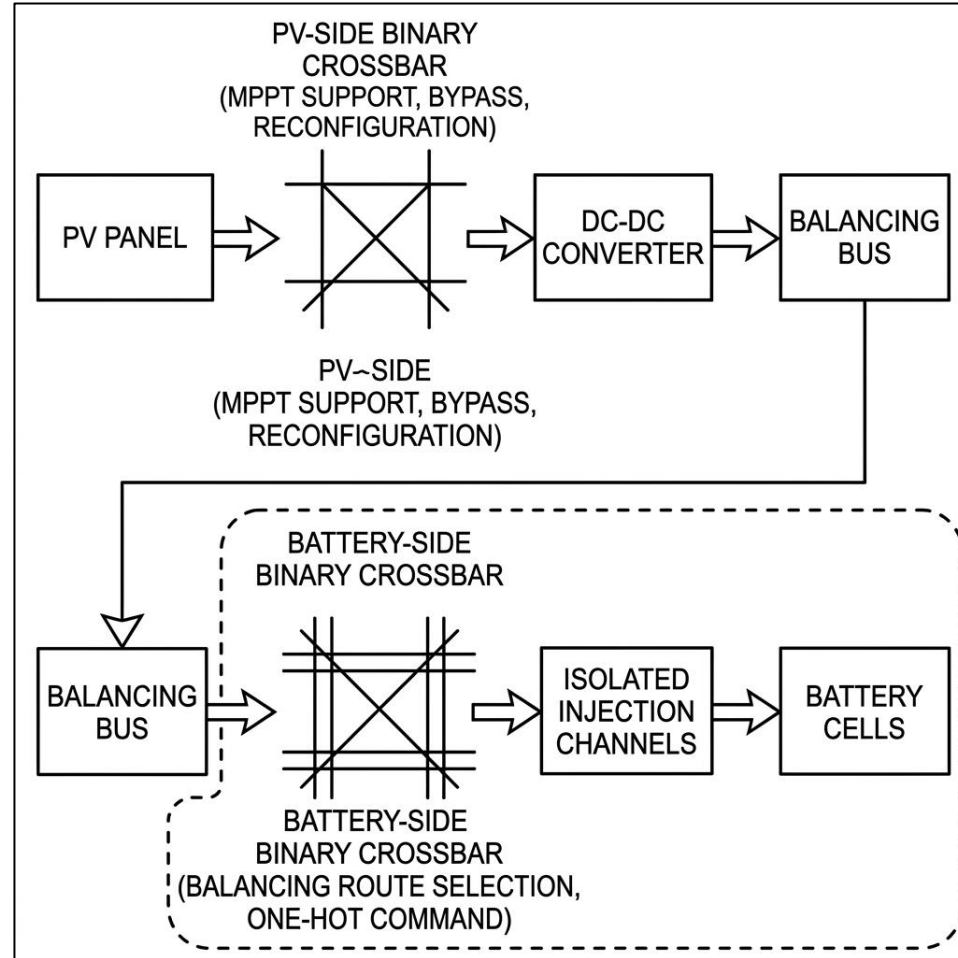
Local reflex modules reduce long analog runs and accelerate balancing correction



Illustrative balancing-time comparison

Why two crossbar Modules are needed

The PV source and the battery pack are physically far apart and solve different problems



PV-side Module

- Near PV cells or segments. It supports bypass, reconfiguration and MPPT stabilization.

Battery-side Module

- Near battery cells or clusters. It selects which weak safe cluster receives conditioned energy.
- The two modules use the same binary crossbar principle, but they are different physical modules with different inputs and outputs.

Capture first on the PV side • Route second on the battery side

System comparison: classical chain vs cooperative crossbar architecture

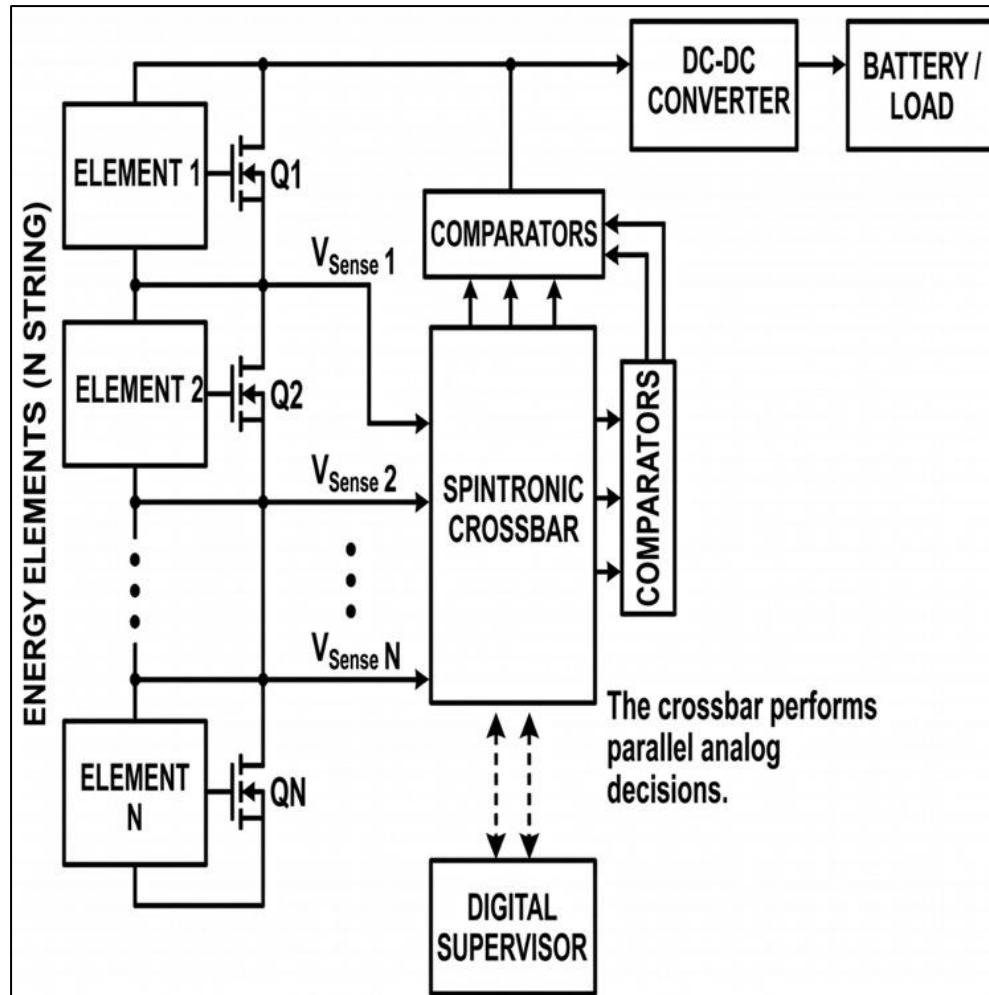
The advantage is system-level: efficiency, speed, wiring simplicity and lower heat

Aspect	Classical PV + BMS	SPIN-PV-BMS cooperative crossbars
PV shading response	Global MPPT sees combined output after the source has changed.	PV-side block commands fast local bypass or reconfiguration.
Battery balancing	Passive burns energy; active adds converter complexity.	Weak safe cells are lifted using PV or auxiliary energy.
Decision latency	Sequential sensing and firmware loops.	Many local binary conditions evaluated in parallel.
Heat and losses	Balancing heat and poor transient operating points.	Lower dissipation and better use of harvested energy.
Wiring and integration	Long analog runs and centralized low-level decisions.	Local modules with short wiring and low-pin event communication.
Safety role	BMS must supervise everything centrally.	BMS remains supervisor; reflex blocks act inside permissions.

Bridge to next part: implementation of the cooperative binary crossbar control modules

Applicability beyond PV: the same reflex principle

PV, TE, TPV and piezoelectric harvesters all need fast local decisions



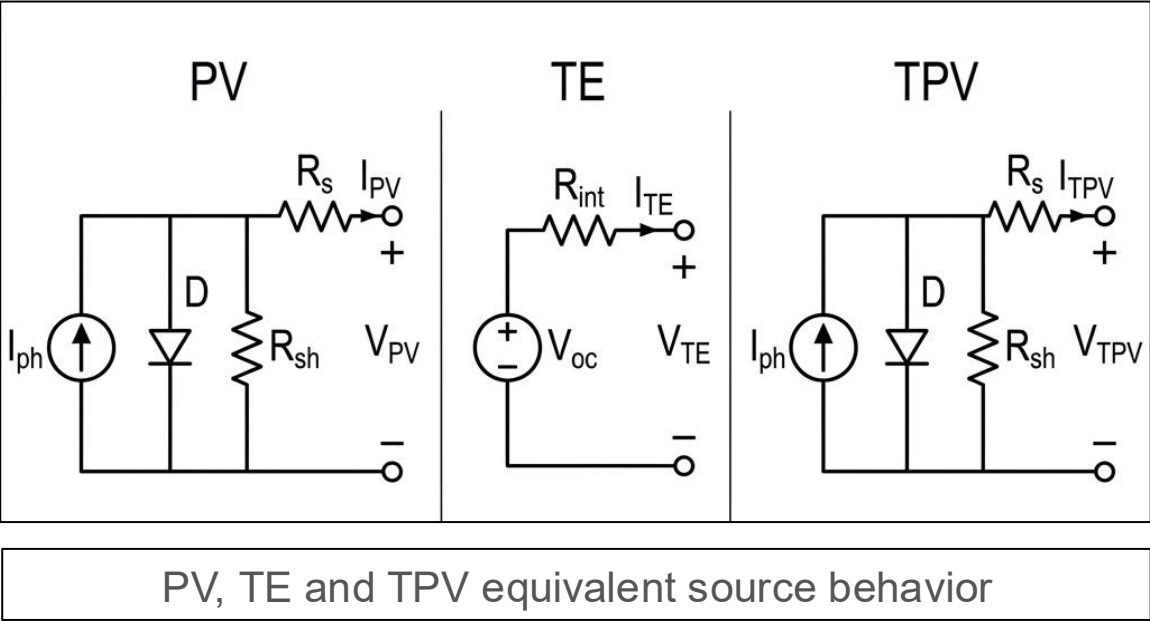
Generalized harvester control architecture

- Not limited to photovoltaic panels.
- Thermoelectric sources: uneven ΔT creates weak modules.
- Thermophotovoltaic cells: non-uniform radiation behaves like PV mismatch.
- Piezo harvesters: vibration amplitude and frequency change locally.
- Auxiliary DC buses can feed the same balancing logic

Different energy sources, same local reflex problem: local mismatch detection and low-energy reconfiguration.

One physical logic across different energy sources

Change the source physics, keep the local decision pattern



Source	Local weak event	Immediate reflex action
PV / TPV	Low irradiance or shaded cell	Bypass / isolate / wake MPPT
TE / TEG	Low ΔT or reverse current risk	Isolate / reconfigure module
Piezo	Detuning, over-voltage, faulty element	Clamp / mode-select / isolate

The crossbar does not need to know the full physics. It only maps local flags into safe local commands.
The architecture is domain agnostic.

EROIE / EROI at small scale: control energy matters

When harvested power is small, the controller must not consume the gain

EROIE view = useful energy delivered / energy spent to capture, decide, convert, route and protect

- At large scale, panel or converter efficiency often dominates.
- At small scale, control energy can dominate
- Sensing, ADCs, firmware loops and wiring losses can become decisive.
- Energy-proportional Physical AI asks for the minimal sufficient decision layer.
- Binary local reflexes are valuable when the rule is simple, repetitive and urgent.

Heavy control

Net useful energy



ADC-light reflex

Net useful energy

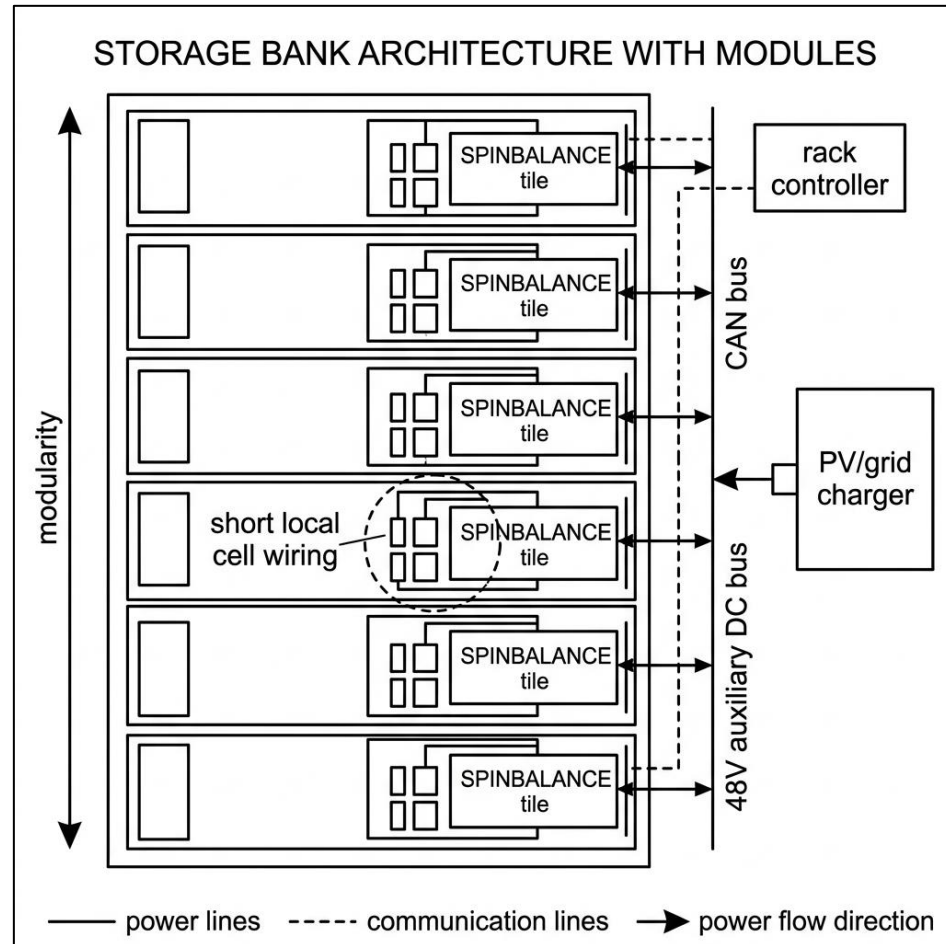


■ Control overhead ■ Useful harvested/stored energy

At the edge, “net useful energy” is the real metric.

Scaling up or down: change granularity, not the logic

From cargo bikes and light EVs to large stationary storage banks

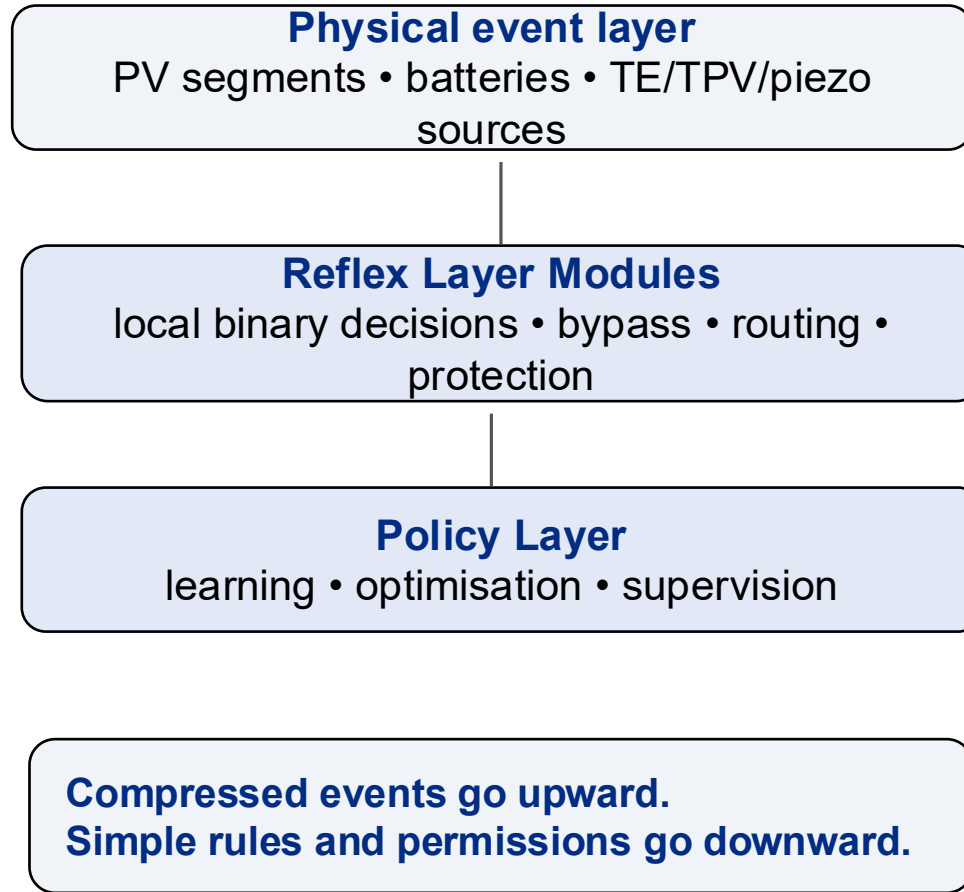


- **Cargo bikes and light EVs:** PV roof, cargo-box PV surface, existing low-voltage service system.
- **Larger vehicles:** operate at module or cluster level to reduce integration burden.
- **Stationary storage:** use rack-level 48 V bus, renewable input or charger.
- **Local modules:** shorter wiring, lower heat, easier service
- **Scalability** comes from choosing the right granularity: cell, cluster, module or rack.

Stationary storage-bank version with local modules

Toward autonomous energy materials and modules

Control intelligence moves closer to the physical event



- Energy devices are not only sources or storage elements. They can become responsive components with local protective and routing intelligence.
- Reflex Layer Modules execute safe local actions at the event location.
- The Policy Layer remains responsible for optimisation, learning and supervision.
- This is not “more computation everywhere”; it is the right physical intelligence in the right place.

Conclusion: from mirror to Physical AI reflex energy control

Binary spintronic crossbars as ADC-light Reflex Layer Modules

Mirror principle

Structure generates a response without central computation.

System problem

Energy harvesting is a full chain: capture, condition, route, store, protect, use.

Binary crossbars

Many decisions are local, binary, repetitive and time-critical.

SPIN-BALANCE vision

Capture energy efficiently; avoid wasting control energy; route energy where it matters most.

Central message: use simple physical reflexes close to the event, and leave learning, optimisation and supervision to the Policy Layer.

Binary spintronic crossbar: compact, non-volatile, local decision substrate, not as a high-precision AI accelerator

Conclusion: Physical AI reflex modules for energy control

Commercial boards today; classical electronics also possible

- This work introduces local, ADC-light reflex modules for urgent energy-state decisions — not an AI accelerator.
- Immediate application: cooperative PV-assisted battery balancing for light EVs and stationary storage.
- Broader opportunity: PV, TE/TPV, piezoelectric harvesters, robotics and IoT edge nodes.
- Demonstrator route: we use commercially available Everspin MRAM/spintronic boards because they are accessible, non-volatile and suitable for fast binary rules.
- Technology note: Everspin/spintronic boards are not strictly necessary. The same reflex logic can also be implemented with more classical electronic manufacturing.
- Expected trade-off: classical electronics may offer slightly lower compactness, standby advantage or speed, but can still deliver gains in PV control, battery balancing and similar local-reflex applications.

Technology acknowledgement

We fully acknowledge that MPPT, bypass switches, active balancing, MRAM and crossbar computing are established fields. Our contribution is not any one of these elements alone. The contribution is the system architecture: using binary spintronic crossbars as non-volatile, ADC-light Reflex-Layer Modules for local energy-state decisions, while classical MPPT and BMS remain the supervisory Policy Layer. **That is our pioneering point.**

Thank you for your attention